DAY 1 • SESSION 3

Artificial Intelligence in Education

Concepts • Tools • Applications

NIELIT Ropar × SCERT Punjab

Artificial Intelligence :
Concepts, Tools and Applications

Session Overview

01 **What is Artificial Intelligence?**
Definition, History & Key Concepts

02 **Types of AI & Machine Learning**
Narrow AI, ML, Deep Learning, Generative AI

03 **AI Tools for Teachers**
ChatGPT, Gemini, Canva AI, Grammarly, QuillBot & more

04 **AI in Education – Applications**
Personalized learning, assessment, content creation

05 **Responsible & Ethical AI Use**
Bias, privacy, academic integrity

What is Artificial Intelligence?

"The science and engineering of making intelligent machines, especially intelligent computer programs that can think, learn, and act like humans."

Learn

Systems that improve from experience without being explicitly programmed

Reason

Analyze data, recognize patterns and make informed decisions

Act

Take actions to achieve goals — from answering questions to driving cars

Quick History: 1950 → 1990 → 2012 → 2022 → Today

1950	1997	2012	2022	2024+
Turing Test	IBM Deep Blue	Deep Learning	ChatGPT	AI Everywhere

Types of AI & Machine Learning

Artificial Intelligence (AI)

Machine Learning (ML) — learns from data without explicit rules

Deep Learning — neural networks inspired by the human brain

Generative AI — creates new content (text, images, audio)

Narrow AI	General AI	Super AI
Designed for ONE specific task (e.g. Face recognition, chess)	Human-level reasoning across all tasks (still theoretical)	Surpasses human intelligence (future concept)

How Does AI Learn?

1 Data Input

Thousands/millions of examples fed to the model

2 Training

Algorithm finds patterns & relationships in data

3 Model

Mathematical representation of learned patterns

4 Prediction

Apply model to new, unseen data

Key Learning Approaches:

Supervised

Learns from labeled data
(e.g. spam vs not spam)

Unsupervised

Finds patterns in unlabeled data
(e.g. customer grouping)

Reinforcement

Learns by reward & penalty
(e.g. game-playing AI)

Popular AI Tools for Teachers

ChatGPT
OpenAI

✦ Lesson planning, Q&A, summaries, quiz generation

Google Gemini
Google

✦ Research assistant, document analysis, explanations

Canva AI
Canva

✦ Auto-generate presentations, posters & worksheets

Grammarly
Grammarly Inc.

✦ Grammar check, writing improvement, tone suggestions

QuillBot
QuillBot

✦ Paraphrasing, summarizing, citation generator

Socratic
Google

✦ Step-by-step homework help for students

🌐 *All tools accessible via browser — No installation required!*

ChatGPT for Teachers - Practical Use Cases

Lesson Planning

Prompt: "Create a 40-min lesson plan for Class 8 on Fractions with activities"

Quiz Generator

Prompt: "Generate 10 MCQs on Photosynthesis for Class 10 with answer key"

Simplify Content

Prompt: "Explain Newton's Laws of Motion in simple language for 12-year-olds"

Emails & Letters

Prompt: "Draft a parent communication regarding upcoming Annual Sports Day"

Translation Help

Prompt: "Translate this English passage to simple Punjabi for students"

Project Ideas

Prompt: "Suggest 5 science project ideas for Class 9 related to environment"

Tip: The more specific your prompt, the better the result!

AI Applications in Education

Personalized Learning

AI adapts content difficulty to each student's level. Khan Academy, Duolingo use this.

Intelligent Tutoring

AI tutors provide 24/7 help — explain concepts, answer doubts, give practice questions.

Auto Grading

Automatically grade MCQs, short answers & essays. Saves teacher time significantly.

Content Creation

Generate reading passages, worksheets, lesson notes & multimedia content instantly.

Learning Analytics

Track student progress, predict at-risk students, identify knowledge gaps.

Accessibility Tools

Text-to-speech, real-time captions, translation — making education inclusive.

AI in Action - Real World Examples

AI Around Us Every Day

- Gmail spam filtering
- Google Search predictions
- Netflix/YouTube recommendations
- Face unlock on smartphones
- Siri, Alexa, Google Assistant
- Face unlock on smartphones
- Google Search predictions
- Netflix/YouTube recommendations
- Google Maps traffic prediction
- Gmail spam filtering
- Auto photo enhancement
- Medical image diagnosis
- Software Development (claude.ai)

IN AI in India

DIKSHA Platform
AI-powered learning for school students

PM e-Vidya
Digital education initiative with AI support

Pratham AI
Personalized reading & math programs

UPI & DigiLocker
AI fraud detection in fintech

NPCI FASTag
AI-based toll & traffic management

CoWIN Platform
AI for vaccine scheduling & analytics

Responsible & Ethical Use of AI

Bias in AI

AI can reflect and amplify human biases present in training data. Always question AI outputs.

Data Privacy

Never share students' personal details with AI tools. Read privacy policies carefully.

Academic Integrity

Using AI to write student assignments = cheating. Teach students to use AI as an aid, not a crutch.

Fact Checking

AI can 'hallucinate' — generate false information confidently. Always verify facts.

Remember: "AI is a tool — YOU are the teacher. Use it wisely!"

Hands-On Activity

Try it yourself - in pairs or individually!

Activity 1 (10 min)

Explore ChatGPT

- Go to chat.openai.com (free)
- Ask: 'Create a lesson plan on Water Cycle for Class 7'
- Edit the plan — is it good? What needs changing?
- Save the output and share with partner

Activity 2 (10 min)

AI Image Generation

- Go to canva.com → Text to Image
- Type: 'A diagram showing solar system for students'
- Download and discuss usefulness in classroom

📌 Lab: 213 (First Floor) | Open browser → No login required for most tools

Key Takeaways

1. AI is transforming every field, including education — teachers must stay ahead.
2. ML, Deep Learning & Generative AI are key sub-fields shaping today's AI tools.
3. Tools like **ChatGPT**, **Gemini** & **Canva AI** can save hours of teacher prep time.
4. AI in education enables personalized learning, smart assessments & analytics.
5. Always verify AI outputs, protect student data, and use AI ethically.

Next: Session 4 — Demonstration / Hands-On Practice | Tea Break: 3:15–3:30 PM

Thank You!

Questions & Discussion

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Teachers Training in ICT & Digital Initiatives

AI is not about replacing teachers — it's about empowering them!

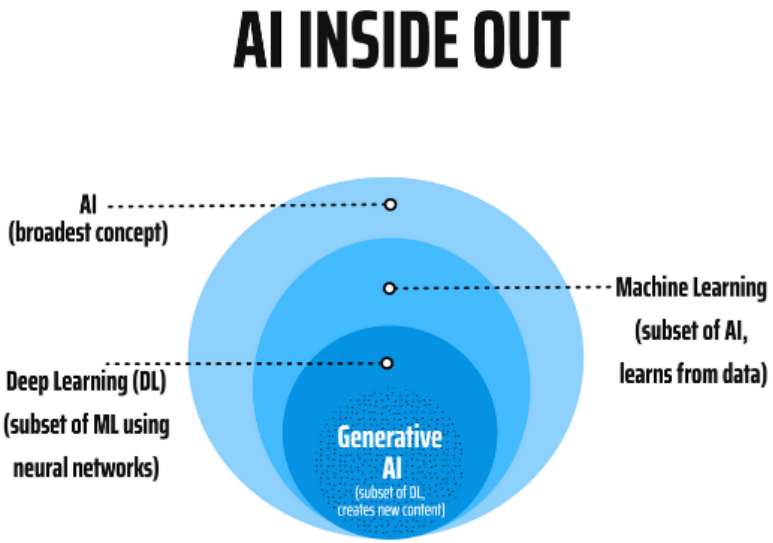
Artificial Intelligence

Foundational understanding and its applications in school education.

What is Artificial Intelligence?

Definition

AI refers to computer systems that can perform tasks typically requiring human intelligence: **Recognizing patterns, solving problems, understanding language, and making decisions.**



Evolution of AI

1950s

Foundations

Turing Test proposed. Early logic-based systems and symbolic AI.

1990s

Data Era

Internet acceleration. Systems began learning from larger datasets (Machine Learning).

2010s

Deep Learning

Cloud computing and neural networks allowed for image and speech recognition.

2023+

Generative AI

Models capable of creating new content: text, code, audio, and images. (ChatGPT)

AI vs. Human Intelligence

Artificial Intelligence

- Extremely fast processing
- Perfect memory (Database)
- Rational & Unbiased (ideally)
- **Narrow focus** (Good at one thing)

Human Intelligence

- Creative & Emotional
- Understand Context
- Adaptable to new situations
- **General intelligence** (Good at many things)

Introduction to Machine Learning (ML)

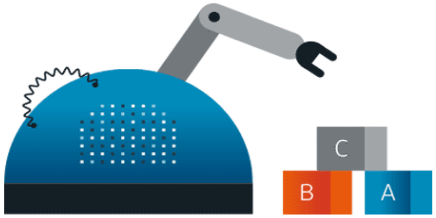
Definition:

Machine learning (ML) is a subset of artificial intelligence (AI) that enables computer systems to learn and improve autonomously from data. Instead of following hard-coded instructions, ML models use statistical algorithms to identify patterns, make predictions, and adapt to new information over time.

ML is the **engine** that makes modern AI work.

The Core Concept

Instead of explicitly programming rules ("If X do Y"), we feed the computer **data** and let it learn the rules itself.



How Machine Learning Works

1. Input

Photos, Text, Numbers

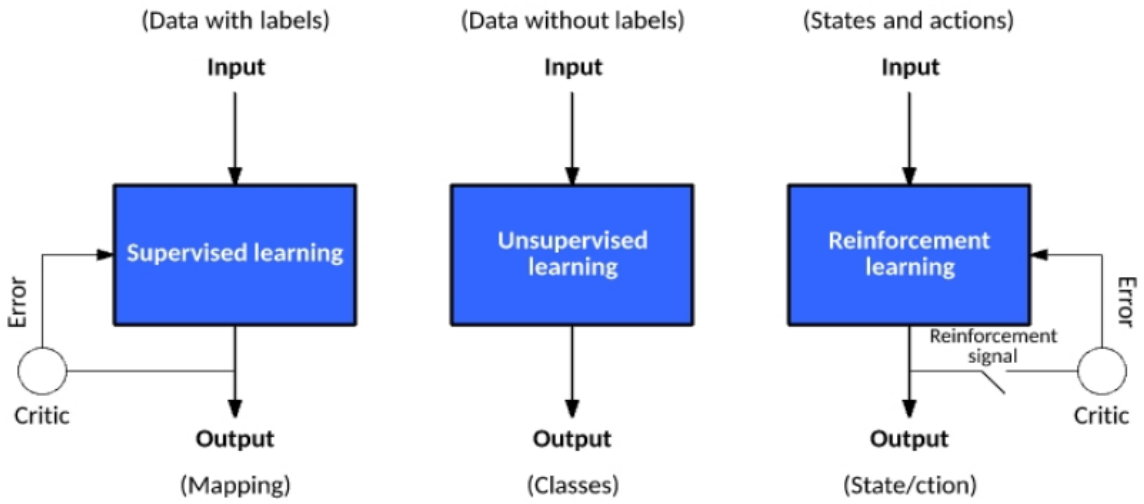
2. Train

Algorithm finds patterns

3. Predict

New Data -> Decision

3 Types of Machine Learning



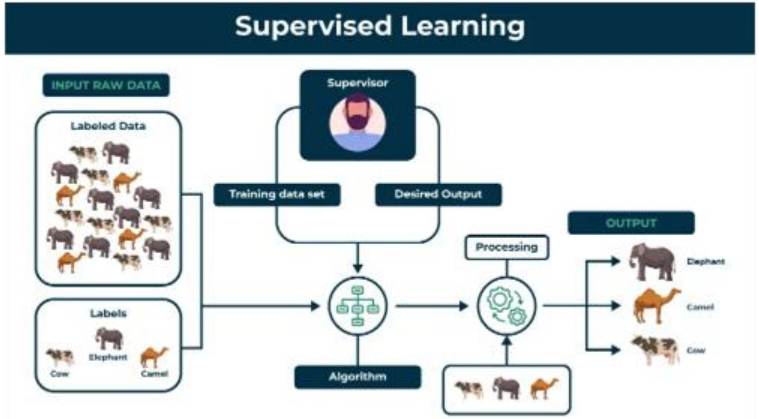
Supervised	Unsupervised	Reinforcement
Task Driven	Data Driven	Reaction Driven

1. Supervised Learning

"Learning with a Teacher"

The model is trained on **Labeled Data**. We show the computer the input AND the correct answer.

Analogy: A teacher showing a child flashcards. "This is an Apple", "This is a Banana". Later, the child identifies them alone.

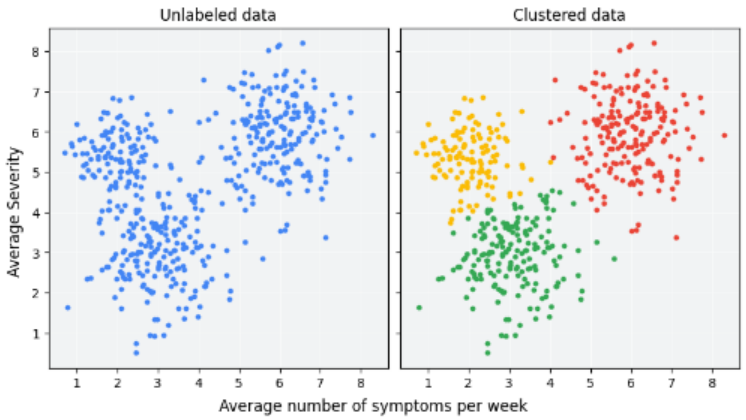


2. Unsupervised Learning

"Self-Learning"

The model gets **Unlabeled Data**. It must find structure, patterns, or groups on its own.

Analogy: Giving a child a bucket of mixed buttons. Without instruction, they might sort them by color, size, or shape.

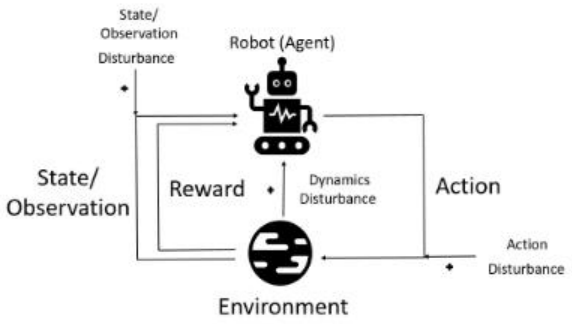


3. Reinforcement Learning

"Trial and Error"

An agent learns to make decisions by performing actions and receiving a **Reward** or **Penalty**.

Analogy: Training a dog. Sit -> Treat (Reward). Jump -> No Treat (Penalty). The dog learns to maximize treats.

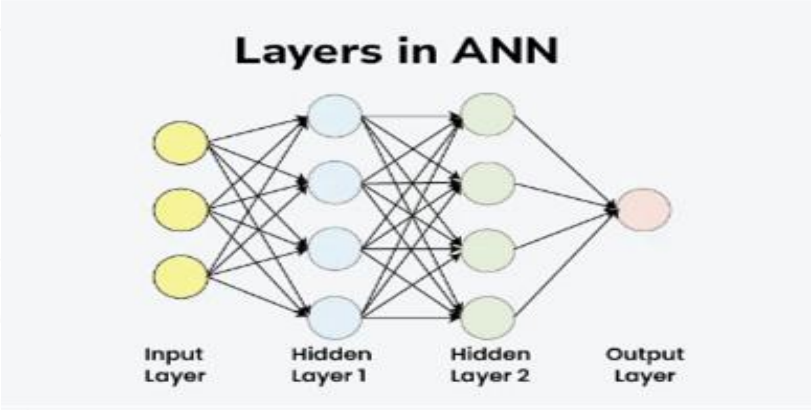


Deep Learning (DL)

Neural Networks

Inspired by the human brain. Uses layers of artificial neurons to process complex, unstructured data like **Images, Audio, and Text**.

- Face Recognition
- Voice Assistants (Alexa/Siri)
- Self-Driving Cars

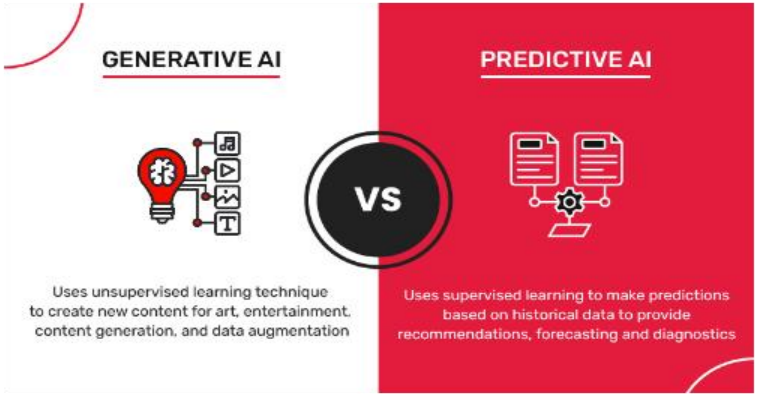


Generative AI (GenAI)

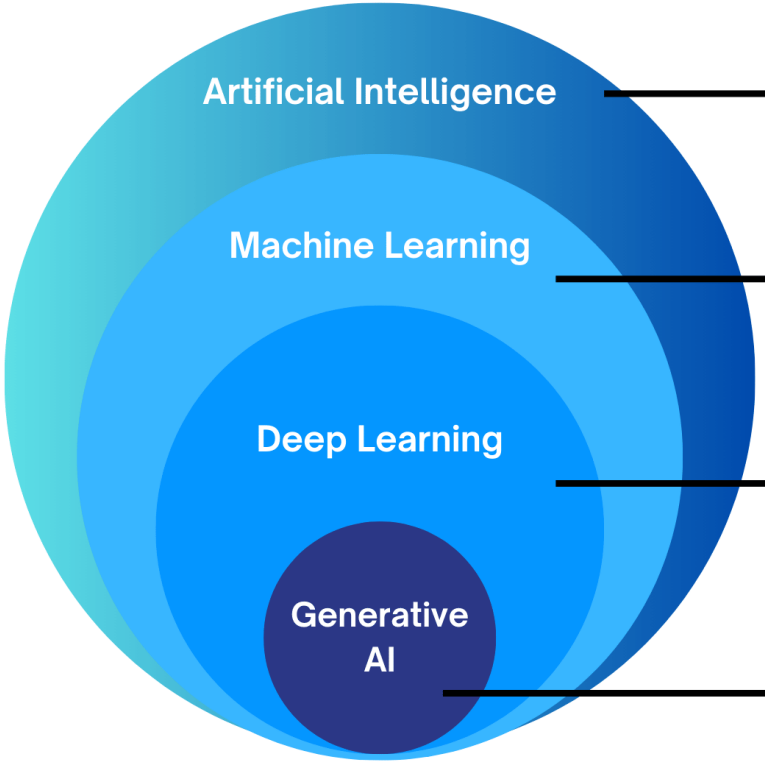
Creation vs. Analysis

While traditional AI analyzes existing data (e.g., "Is this a cat?"), Generative AI creates **NEW** data (e.g., "Draw a cat in space").

Tools: ChatGPT/Claude (Text), Midjourney (Images), Suno (Music).



The AI Hierarchy



AI - machines that mimic human intelligence to perform tasks like problem-solving and decision-making.

ML - machines learn from data to improve their performance without explicit programming.

Using Neural Networks with multiple layers to analyze data, inspired by the human brain's structure.

AI that creates new content, like text, images, or music, based on learned patterns from data.

Live Project: Diabetes Prediction

Type: Supervised Machine Learning (Classification)

Live demo: https://nielitropar-diabetes.hf.space (Classification)

Input Data

Glucose, BMI, Age, Insulin Level

Algorithm

Decision Tree / Logistic Regression

Output

Risk: **Positive / Negative**

Live Project: Sentiment Analysis

Type: Deep Learning (NLP / Computer Vision)

Live demo: https://nielit-ropar.streamlit.app (Computer Vision)

Input

Student Facial Expression or Text Feedback

Neural Net

Analyzes features (eyebrows, mouth shape)

Output

Emotion: **Happy, Confused, Bored**

AI Applications in School

Administration

- Automated Attendance (Face Rec)
- Grading Objective Tests
- Timetable Scheduling

Pedagogy

- Adaptive Learning (Personalized)
- Intelligent Tutoring Systems
- Real-time Language Translation

AI Solutions for Key Educational Challenges.

Tool	Function
MagicSchool.ai	Lesson Planning, IEP Generation, Rubrics
Canva Magic Write	Creating Visual Presentations & Worksheets
ChatGPT / Gemini / claude.ai	Drafting emails, summarizing text, brainstorming
Curipod	Interactive AI-powered slide lessons

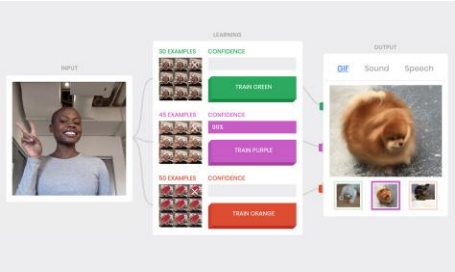
Note: Ensure these tools are used ethically and responsibly.

Source: https://ditchthattextbook.com/ai-tools/

Tools for Students

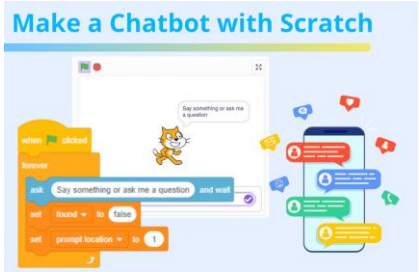
Teachable Machine

Train your own ML model with a webcam. No code required.



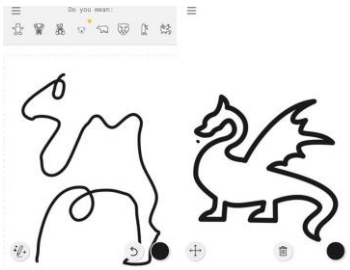
Scratch AI

Block coding with AI extensions (Face sensing, Speech-to-text).



AutoDraw

Turns rough sketches into polished icons using AI.



Limitations & Ethics

Algorithmic Bias

AI reflects the biases in its training data (e.g., gender or racial bias).

Hallucinations

GenAI can make up facts confidently. Always verify the source.

Data Privacy

Never share sensitive student information (Names, IDs) with public AI tools.